

Alonso Marcos Muñoz

Data Engineer

Albacete, Spain | alonsomarcosm99@gmail.com

linkedin.com/in/alonsomarcosm99 | github.com/alonsomarcosm99 | alonsomarcosm99.github.io/en

SUMMARY

Data Engineer with almost 2 years of experience in ETL pipelines, PostgreSQL and Oracle modelling and production deployment. Part of the ImpulsaDATA platform team at Tragsatec (15 people in a project of around 80), serving 22 Spanish ministries and public bodies. MSc in Big Data and Cloud Computing with hands-on Databricks, Spark, Kafka, Airflow and AWS.

EXPERIENCE

Tragsatec

Albacete, Spain

Data Engineer

Oct. 2025 - Present

- Build and maintain Python and SQL ETL pipelines that collect, validate and publish ImpulsaDATA catalogues daily or weekly, in a programme that has released 5,771 open datasets.
- Design PostgreSQL and Oracle transformations, aggregations and views so that data arrives structured, consistent and ready to be queried and published.
- Automate the checks that validate every dataset against the European open data standard before release, keeping errors out of official catalogues.
- Set up and run development, test, pre-production and production environments with Docker, Podman, Git and SSH, keeping deployments documented, repeatable and easy for the team to review.
- Within the AgoraData project, lead development of the web interface used to operate ImpulsaDATA's backend and Python processes, built with Java, Spring Boot, Vaadin and PostgreSQL.
- Coordinate and delegate security, testing, accessibility and CI/CD for the AgoraData project interface across the team, so work moves forward in parallel.

Tragsatec

Albacete, Spain

Data Engineer (Internship)

Jun. 2025 - Sep. 2025

- Developed the first version of the project's metadata conversion tool, a Python ETL that automated a previously manual flow and publishes the information in CKAN.

La Fábrica del Tiempo

Albacete, Spain

Data & Automation Engineer (Internship)

Feb. 2024 - Aug. 2024

- Improved the efficiency of lead management, notifications and approvals by more than 25%, automating them with Power Automate across SharePoint, the CRM and Microsoft 365.
- Delivered an internal Power Apps application that centralises the company's process management and information for a team of 10+ people.

PROJECTS

- **Databricks Telco Churn Lakehouse:** Co-built an end-to-end data platform that predicts telecom customer churn from 16 million synthetic records of 2 million customers, using PySpark, Delta Lake, Unity Catalog and MLflow: <https://github.com/alonsomarcosm99/databricks-telco-churn-lakehouse>
- **Batch and streaming platform:** Implemented a batch and real-time data platform with Spark, Kafka and Airflow that organises data into layers of increasing quality (Medallion architecture) on Delta Lake: <https://github.com/alonsomarcosm99/spark-kafka-airflow-data-platform>
- **AWS Smart Parking:** Deployed an AWS IoT platform that delivers data from 40 simulated parking sensors to its API in under 2 seconds at an estimated cost below USD 5 per month, using Lambda, DynamoDB and MQTT: <https://github.com/alonsomarcosm99/smart-parking-albacete>

EDUCATION AND CERTIFICATIONS

UCLM - MSc Big Data and Cloud Computing

2025 - 2026

- Spark, Databricks, streaming, AWS, machine learning, IoT and data governance. Master's thesis graded 9.2/10.

UCLM - BSc Computer Engineering, Information Technology

2019 - 2025

IES Juan Bosco - Higher Diploma in Networked Systems Administration

2017 - 2019

CAPM, project management certification - Project Management Institute

Valid until 2028

B2 First - Cambridge English

2025

SKILLS

Data Engineering: Python, SQL, ETL, data modelling, PostgreSQL, Oracle, REST APIs, data quality and traceability.

Platforms and governance: CKAN, OpenMetadata, DCAT-AP-ES, RDF, JSON-LD, SHACL, Power Apps and Power Automate.

Applied Big Data and cloud: Databricks, PySpark, Spark, Delta Lake, Airflow, Kafka, MLflow, AWS and Terraform.

Development and operations: Java, Spring Boot, Vaadin, Docker, Podman, GitLab CI/CD, Git, Linux and SSH.

Languages: native Spanish and professional English (B2 First).